LIBRARY MANAGEMENT SYSTEM

CHANGE REQUEST SPECIFICATION REPORT

CR001 • CR002 • CR003 • CR004

Document Field	Details
System	Library Management System
Change Request IDs	CR001, CR002, CR003, CR004
Document Type	Change Request Specification
Baseline	Existing Library Management System
Status	Proposed / Pending Implementation and Testing

1. Introduction

This report consolidates four requested enhancements for the existing Library Management System. Each change request describes the business requirement, current behaviour, requested change, scope, potentially affected files, functional requirements, acceptance criteria, test cases and expected outcome. The report specifies the requested changes; it does not claim that they have already been implemented or tested.

CR001: Student Login and Personal Borrowings / Holds

Field	Details
Change Request ID	CR001
Title	Student Login and Personal Borrowings / Holds
Priority	High
Difficulty	Medium
Change Type	Frontend and Backend Enhancement
Affected Version	Existing Library Management System baseline
Dependency	Existing student records, authentication, borrowing and hold modules

2. Business Requirement

Students should be able to log in to the Library Management System using their own credentials and view their personal borrowing records and holds. This request adds student-facing access that was previously identified as out of scope in the README. Students must only be able to access information associated with their own account.

3. Current Behaviour

- The README identifies student login and student self-service access to borrowings and holds as out of scope.
- The requested change requires a student authentication flow and a personal view of borrowing and hold records.
- Existing administrator and library-management functionality should continue to work.

4. Requested Change

- Provide a student login interface.
- Authenticate student credentials securely and create an authenticated student session.
- Provide a student dashboard or account page showing the student's own borrowing records.
- Display the student's own holds, including relevant status information where available.
- Enforce authorization on the server so a student cannot view another student's records.
- Provide clear validation and error messages for invalid login attempts.
- Preserve existing administrator access and permissions.

5. Scope

In Scope

- Student login interface and authentication flow.
- Student-only dashboard or account page.
- Display of the logged-in student's borrowings and holds.
- Server-side authorization and session handling.
- Validation and login error handling.

Out of Scope

- Changes to unrelated library workflows.
- Changes to administrator roles beyond what is required to protect access.
- New borrowing or hold business rules not specified in this request.

6. Expected / Potentially Affected Files

File / Area	Reason
README.md	Update scope documentation if the feature is approved and implemented.
login page / authentication files	Provide student login and authentication.
student dashboard or account page	Display personal borrowing and hold records.
borrowings module / query files	Retrieve borrowings for the authenticated student.

holds module / query files	Retrieve holds for the authenticated student.
authorization middleware / session handling	Restrict records to the logged-in student.
style.css or relevant stylesheet	Style login and student account screens.

7. Functional and Security Requirements

- FR-001: The system shall provide a student login interface.
- FR-002: The system shall validate student login credentials.
- FR-003: An authenticated student shall be able to view their own borrowing records.
- FR-004: An authenticated student shall be able to view their own holds.
- SEC-001: Students shall not be able to access another student's borrowings or holds.
- SEC-002: Protected pages and endpoints shall require authentication.
- SEC-003: Existing administrator permissions shall remain protected.

8. Acceptance Criteria

1. A student can open the login page and submit credentials.
2. Valid credentials allow the student to access the student account area.
3. Invalid credentials are rejected with an appropriate message.
4. The borrowing list contains only records belonging to the logged-in student.
5. The holds list contains only holds belonging to the logged-in student.
6. Unauthenticated users cannot access student-only pages or endpoints.
7. Existing administrator functionality continues to operate.

9. Test Cases

Test ID	Scenario	Expected Result	Status
TC-001	Open student login page	Login form is displayed	Pending
TC-002	Log in with valid student credentials	Student account page opens	Pending
TC-003	Log in with invalid credentials	Access is denied and error is shown	Pending
TC-004	View personal borrowings	Only the logged-in student's records appear	Pending
TC-005	View personal holds	Only the logged-in student's holds appear	Pending
TC-006	Attempt to access another student's records	Access is denied	Pending
TC-007	Open protected page while logged out	User is redirected or denied	Pending

CR002: Fine Calculation for Overdue Borrowings

Field	Details
Change Request ID	CR002
Title	Fine Calculation for Overdue Borrowings
Priority	High
Difficulty	Medium
Change Type	Database, Backend and Reporting Enhancement
Affected Version	Existing Library Management System baseline
Dependency	Borrowing records, due dates, return workflow and reports

2. Business Requirement

The system should calculate a fine for an overdue book using the number of overdue days multiplied by the configured per-day fine rate. When a book is returned, the calculated fine should be displayed to the user or staff member processing the return. Fine information should also be available in relevant reports.

3. Current Behaviour

- The requested change identifies the need to add or complete fine calculation for overdue borrowings.
- The database schema may need fine-related columns.
- The return workflow and reports need to display the calculated fine.

4. Requested Change

- Add the required fine fields to schema.sql, subject to the existing database design.
- Calculate overdue days from the due date and the actual return date.
- Calculate fine as overdue days multiplied by the per-day fine rate.
- Do not calculate a late fine when the book is returned on or before the due date.
- Display the calculated fine during the return process.
- Persist the calculated fine where required and include it in relevant reports.
- Handle date boundaries consistently and avoid negative fine amounts.

5. Scope

In Scope

- Fine-related database fields or schema changes.
- Overdue-day and fine calculation logic.
- Fine display in the return workflow.
- Fine details in relevant reports.

Out of Scope

- Online fine payment integration unless separately approved.
- Changes to unrelated borrowing policies.
- Waivers, discounts or payment reconciliation rules not specified in this request.

6. Expected / Potentially Affected Files

File	Reason
schema.sql	Add or update fine-related fields and appropriate defaults.
return.php	Calculate and display fine when a book is returned.
reports	Include fine amounts and overdue information in relevant reports.

7. Functional Requirements

- FR-001: The system shall determine whether a borrowing is overdue.
- FR-002: The system shall calculate overdue days using the due date and return date.
- FR-003: The fine shall be calculated as overdue days × per-day fine rate.
- FR-004: The return workflow shall display the calculated fine.
- FR-005: Fine information shall be available in relevant reports.
- FR-006: Fine amounts shall not be negative.

8. Acceptance Criteria

8. A book returned on or before its due date has no overdue fine.
9. A book returned after its due date has its overdue days calculated correctly.
10. The fine equals overdue days multiplied by the configured daily rate.
11. The return screen displays the calculated fine before or during completion of the return.
12. The fine is stored or retained according to the database design.
13. Relevant reports display the fine amount accurately.
14. Normal return processing continues to work for both overdue and non-overdue books.

9. Test Cases

Test ID	Scenario	Expected Result	Status
TC-001	Return book on due date	Fine is 0	Pending
TC-002	Return book before due date	Fine is 0	Pending
TC-003	Return book 1 day overdue	Fine equals one daily rate	Pending
TC-004	Return book multiple days overdue	Fine equals overdue days × daily rate	Pending
TC-005	Check return screen	Calculated fine is displayed	Pending
TC-006	Check relevant report	Fine amount is shown correctly	Pending

CR003: Borrowings Search and Filter

Field	Details
Change Request ID	CR003
Title	Borrowings Search and Filter
Priority	Medium
Difficulty	Low
Change Type	Frontend and Backend Enhancement
Affected Version	Existing Library Management System baseline
Dependency	Borrowings listing and existing borrowing records

2. Business Requirement

Staff should be able to find borrowing records more efficiently by searching with a student name or book and filtering records by borrowing status. The supported statuses are Issued and Returned.

3. Current Behaviour

- The borrowing list needs search and filter controls to narrow down records.
- The specified implementation location is borrowings/index.php.

4. Requested Change

- Add a search input for student name and book title or name.
- Add a status filter with Issued and Returned options.
- Allow search and status filtering to be used together.
- Show an appropriate message when no records match the criteria.
- Preserve existing borrowing list and record actions.

5. Scope

In Scope

- Search by student name.
- Search by book name or title.
- Filter by Issued or Returned status.
- Display matching borrowing records and an empty-results message.

Out of Scope

- Pagination unless separately requested.
- Advanced analytics or unrelated report changes.
- Changes to borrowing or return rules.

6. Expected / Potentially Affected Files

File	Reason
borrowings/index.php	Add search and status filter controls and show filtered results.

7. Functional Requirements

- FR-001: The borrowing page shall allow search by student name.
- FR-002: The borrowing page shall allow search by book title or name.
- FR-003: The borrowing page shall filter by Issued status.
- FR-004: The borrowing page shall filter by Returned status.
- FR-005: Search and status filtering shall work together.

- FR-006: The system shall display a clear message when no records match.

8. Acceptance Criteria

15. Searching a student name displays matching borrowing records.
16. Searching a book title displays matching borrowing records.
17. Selecting Issued displays only issued records.
18. Selecting Returned displays only returned records.
19. Search text and status filter can be applied together.
20. Clearing search and filter controls restores the full list.
21. No-match searches display a helpful empty-state message.

9. Test Cases

Test ID	Scenario	Expected Result	Status
TC-001	Search by student name	Matching records are shown	Pending
TC-002	Search by book title	Matching records are shown	Pending
TC-003	Filter status as Issued	Only issued records are shown	Pending
TC-004	Filter status as Returned	Only returned records are shown	Pending
TC-005	Combine search and status	Only records matching both criteria are shown	Pending
TC-006	Search with no matches	Empty-results message is shown	Pending

CR004: Overdue Borrowing Highlighting and Badge

Field	Details
Change Request ID	CR004
Title	Overdue Borrowing Highlighting and Badge
Priority	Medium
Difficulty	Low
Change Type	Frontend Enhancement
Affected Version	Existing Library Management System baseline
Dependency	Borrowings list, due-date data and stylesheet

2. Business Requirement

The borrowing list should make overdue records easy to identify. If a borrowing's due date has passed and the book has not been returned, the record should be highlighted in red and display an “Overdue” badge.

3. Current Behaviour

- The requested change identifies the need for stronger visual indication of overdue records.
- The specified implementation locations are borrowings/index.php and style.css.

4. Requested Change

- Compare the due date with the current date for records that have not been returned.
- Highlight overdue borrowing rows or relevant record details in red.
- Display an “Overdue” badge for each overdue, unreturned record.
- Do not mark returned records as overdue merely because their due date is in the past.
- Keep the highlighting readable and consistent with the existing interface.

5. Scope

In Scope

- Detect overdue unreturned borrowing records.
- Red highlighting for overdue records.
- An “Overdue” badge.
- CSS styling for the visual indicator.

Out of Scope

- Changing due dates or borrowing policies.
- Automatic notifications or email reminders.
- Fine calculation itself, which is covered by CR002.

6. Expected / Potentially Affected Files

File	Reason
borrowings/index.php	Determine overdue status and render the badge/highlighting.
style.css	Define red highlighting and badge appearance.

7. Functional Requirements

- FR-001: The system shall identify unreturned borrowings whose due date is before the current date.
- FR-002: Overdue records shall be highlighted in red.
- FR-003: Overdue records shall display an “Overdue” badge.
- FR-004: Returned records shall not be shown as currently overdue.

- FR-005: Visual indicators shall remain legible and consistent with the existing layout.

8. Acceptance Criteria

22. An unreturned record past its due date is highlighted in red.
23. An “Overdue” badge is visible for each overdue, unreturned record.
24. A record not yet past its due date is not highlighted as overdue.
25. A returned record is not labelled as currently overdue.
26. The visual changes do not break the borrowing list layout.

9. Test Cases

Test ID	Scenario	Expected Result	Status
TC-001	Unreturned record with past due date	Red highlight and Overdue badge appear	Pending
TC-002	Unreturned record with future due date	No overdue highlighting or badge	Pending
TC-003	Returned record with past due date	Not shown as currently overdue	Pending
TC-004	View borrowing list at normal screen size	Layout remains readable	Pending

10. Implementation and Verification Notes

The file paths listed in this report are the paths identified in the change requests supplied by the requester. They should be checked against the uploaded ZIP structure before development. Any additional files changed during implementation should be documented with a brief reason.

All test cases are marked Pending because no implementation or test evidence was provided for this report. After development, update each status to Pass or Fail and attach screenshots or other evidence. A change request should be marked successfully implemented only after its acceptance criteria have been tested.

11. Final Status

Item	Status
CR001	Proposed / Pending implementation and testing
CR002	Proposed / Pending implementation and testing
CR003	Proposed / Pending implementation and testing
CR004	Proposed / Pending implementation and testing
Outstanding Issues	To be updated after implementation and testing